Predictive Multi-Criteria Allocation across DeFi Lending Protocols:
A Forecast-Driven Extension of the AI-Managed ERC-4626 Vault on the `fractal-defi` Substrate

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Abstract

We study dynamic capital allocation between Aave V3 and Compound V3 USDC supply markets on Ethereum mainnet and ask whether replacing a reactive EMA-smoothed rate observation with a 12-hour-ahead learned forecast yields a statistically and economically significant improvement in risk-adjusted return. The proposed allocator combines a dual-branch DA-BiGRU-CNN forecaster — decomposing each protocol's supply rate into a deterministic kink component and a learned residual — with a four-factor TOPSIS-style multi-criteria decision rule inherited from the author's prior ERC-4626 yield vault [19], and is implemented as a concrete subclass of a new `BaseLendingAllocationStrategy` abstraction contributed to the `fractal-defi` research substrate of [12, 13]. Five strategies (buy-and-hold, APY-only greedy, reactive MCDM-EMA, CIR-calibrated MCDM, and the proposed predictive MCDM) are backtested over an 18-month historical window (Nov 2024 – Apr 2026) with a 10/4/4-month train/validation/test split, and fifteen ablations isolate the contribution of each design choice. The forecaster design is the lending-rate analogue of the dual-branch limit-order-book mid-price predictor of [20], with kink subtraction encoding a hard domain prior unavailable in the LOB setting. Forward-looking targets, set prior to test-set exposure, are an out-of-sample R^2 of 0.15–0.40 for the 12-hour rate level, 55–65% directional accuracy on the "Aave > Compound at $t + 12$h" label, and a Sharpe-ratio improvement of 0.2–0.5 over the reactive-EMA baseline; the null hypothesis that the forecast adds no value over EMA is entertained explicitly and is structurally falsifiable by the design.

1 Introduction

1.1 Motivation

Decentralized lending protocols offer USDC supply-side yield that fluctuates with utilization, market demand, and protocol-specific governance parameters. Supply rates on Aave V3 and Compound V3 are not synchronized: they cross repeatedly over time, and [7] establish empirically that Compound borrowing rates *lead* Aave, with cointegration coefficients implying Aave adjusts at speed 0.607 toward the long-run Compound relationship. A capital provider who can anticipate these crossovers and rebalance ahead of them captures a yield differential that a reactive observer misses. Yet the production-grade DeFi yield aggregators surveyed in Section 2 all act on observed (or EMA-smoothed) rates, and the public research literature on `fractal-defi`-based backtesting [12, 21, 22] covers AMM liquidity provision and basis trading, but not the *lending* microstructure panel of the same substrate. This paper fills that gap.

*Code and reproducibility artifacts: https://github.com/SergeySolovyev/predictive-mcdm-defi. This work realizes the methodological transfer outlined in Section 6.4 of [20] and previewed at the Scientific Telegraph Young Researcher Conference, Central University, Moscow, 17 May 2026.

The forecasting problem is structurally analogous to high-frequency mid-price prediction in limit-order books: noisy multivariate time series, regime-dependent dynamics, a weighted-error metric in which large moves matter disproportionately, and a discrete decision rule (rebalance or hold). The methodological transfer from the author's LOB-prediction work [20] to the present DeFi-rate setting is the central scientific claim. The transfer is not mechanical: a hard domain prior (the protocol-known piecewise-linear "kink" rate function) is available in DeFi and absent in LOBs, and the architecture exploits it via residual subtraction.

1.2 Contributions

1. **Predictive MCDM allocator.** A 12-hour-ahead forecast replaces the reactive EMA observation in the four-factor TOPSIS-style MCDM rule of [19], producing the first published forecast-driven multi-criteria USDC allocator across Aave V3 and Compound V3 on mainnet historicals.
2. **Dual-branch forecaster with kink subtraction.** A DA-BiGRU-CNN architecture predicts each protocol's utilization and rate residual separately, then reconstructs the rate via the protocol-known kink function. This is the lending-rate analogue of the dual-branch LOB predictor of [20], with the kink prior as a genuine architectural contribution beyond either prior work.
3. **Composite forecast loss.** A weighted combination of MSE, weighted-Pearson correlation, and a quantile-90 term aligns the training objective with allocation-relevant accuracy (large moves and the upper tail of the rate distribution).
4. **Open-source contributions to `fractal-defi`.** Two pull requests of independent utility: a Compound V3 lending history loader with a uniform `utilization` field exposed on both Aave and Compound `GlobalState` objects (Extra+1), and a `BaseLendingAllocationStrategy` abstraction generalizing the multi-protocol allocator pattern (Extra+2).
5. **Honest negative-result protocol.** The null hypothesis (forecast adds no value over EMA on this horizon) is operationalized, pre-registered, and entertained as a publishable outcome; the fifteen-row ablation matrix is structured to attribute any observed effect to forecast, criterion structure, or protocol-pair selection.

1.3 Paper Outline

Section 2 reviews lending mechanics, MCDM in financial allocation, deep-learning rate forecasting, the bridge to LOB microstructure, and the recent "fair interest rates are impossible" counter-result [8]. Section 3 situates the work on the `fractal-defi` substrate and describes the new `BaseLendingAllocationStrategy` abstraction. Section 4 develops the dual-branch forecaster and decision rule. Section 5 and Section 6 cover the 18-month data window and hyperparameter calibration. Section 7 states the backtesting protocol; Section 8 hosts the headline and per-baseline tables (with numerical entries deferred to Week 3 execution). Section 9 interprets the findings and addresses the Halpern–Pass–Saraf objection. Section 10 closes with limitations and future work.

2 Background and Related Work

2.1 DeFi Lending Mechanics: Aave V3 and Compound V3

Both Aave V3 and Compound V3 (Comet) implement piecewise-linear *kink* rate models: borrow APY rises linearly with utilization $u \in [0, 1]$ at slope s_1 up to an "optimal" utilization u^*, then steeply at slope s_2. Supply APY is derived from borrow APY, utilization, and a protocol-fixed reserve factor. Despite identical functional form, the two protocols differ in parameter calibration, governance cadence, and oracle architecture; the result is short-horizon dispersion in observed USDC supply rates with periodic crossovers, as analyzed empirically in [3, 7]. The reserve-factor adjustment on either side is not learned from data: it is published on-chain. This makes the kink function an ideal hard prior for a forecast architecture (Section 4).

2.2 MCDM in Financial Allocation

Multi-criteria decision making originated in operations research with SAW (simple additive weighting), TOPSIS [11], AHP [16], and PROMETHEE [5]. Recent applications to cryptocurrency portfolio allocation include [1] and the asymmetric-PROMETHEE-II hybrid of [10]. The author's prior vault paper [19] adopts a four-factor weighted-sum aggregator (APY, risk, cost, stability) with default weights $(w_1, \ldots, w_4) = (0.40, 0.25, 0.20, 0.15)$ verified end-to-end on testnet. The present work inherits this aggregator unchanged and replaces the APY input alone with a forecast, isolating the contribution of prediction from criterion design.

2.3 Deep-Learning Interest-Rate Forecasting

Three recent works directly motivate the deep-learning treatment of DeFi rates. [2] demonstrates 3-hour-resolution adaptive control of Compound rates, implying that 12-hour-resolution forecasting has at least as much signal. [4] formulates DeFi interest-rate adjustment as reinforcement learning on a multi-year Aave dataset, including the March 2023 USDC depeg as a stress regime that we re-use for out-of-distribution testing (Section 8, ablation 14). [25] treats governance parameters as the action space. Classical baselines descend from the Vasicek [23] and CIR [6] short-rate models and the data-driven partitioning extension of [15], which we adapt as Baseline D.

2.4 Bridge to Limit-Order-Book Microstructure

The dual-branch architecture of the author's LOB paper [20] predicts mid-price moves from order-flow and volume streams via late fusion, in the lineage of [17] and the DeepLOB family [18, 24, 26]. The structural similarity to DeFi-rate prediction (multivariate noisy series, heteroscedastic moves, ranking-relevant evaluation) is the methodological foundation of the present work. The principal difference is the availability of the deterministic kink function, which we exploit via residual subtraction (Section 4.1).

2.5 The "Fair Interest Rates Are Impossible" Objection

[8] prove, by reduction of dynamic-repayment lending to barrier-option pricing, that no fair interest rate exists in a fully general lending pool with strategic borrowers. We address this head-on. Our claim is operational, not equilibrium-theoretic: we forecast on a 12-hour horizon and exploit transient cross-protocol dispersion, which is orthogonal to the long-run existence of a single fair rate. Section 9.3 develops this argument in full.

3 System Architecture

3.1 The `fractal-defi` Substrate

We build on `fractal-defi` [13], the Python research library underlying [12] and the liquidity-provision papers [21, 22]. The library exposes a typed entity–strategy contract: a `BaseLendingEntity` encapsulates per-protocol state (`lending_rate`, `borrowing_rate`, balances, gas context), a `BaseStrategy` consumes the joint state and emits typed `ActionToTake` instructions, and a backtest runner sequences entities and strategy under a deterministic execution contract. We

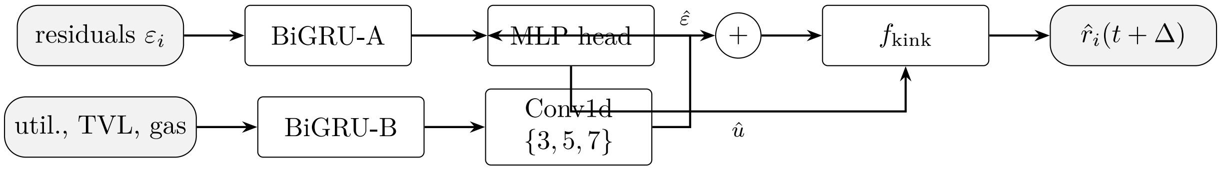

Figure 1: Dual-branch forecaster with kink subtraction. Branch A predicts the rate residual after removing the deterministic kink contribution; Branch B predicts utilization; late fusion reconstructs the rate via the protocol-known f_{kink}.

pin `fractal-defi==1.3.2`; the project's sign-convention lock-in test asserts borrowing_rate $\geq$ lending_rate at every timestamp, guarding against the pre-v1.4.0 sign-flip bug.

3.2 Entity–Strategy Abstraction

Two lending entities are instantiated: the upstream `AaveEntity` and a new `CompoundV3Entity` introduced by this work (Section A). Both expose a uniform `utilization` field on their `GlobalState`, computed from `totalLiquidity` and `totalCurrentVariableDebt` in the subgraph response. The forecaster is loaded as an ONNX module inside the strategy and consumes the joint state via a rolling buffer of 168 hourly observations.

3.3 `BaseLendingAllocationStrategy`: the Extra+2 Contribution

Although `fractal-defi` ships with four lending entities, no multi-lending-protocol allocator was provided at the time of this work. We contribute `BaseLendingAllocationStrategy`, an abstract template exposing four overridable hooks — criterion vector computation, criterion aggregation, target selection, and the rebalance gate — that any future allocator can subclass. The proposed `PredictiveMCDMStrategy` is the first concrete instance; Baselines B, C, and D (Section 7) are implemented as further instances, ensuring apples-to-apples comparison.

4 Methodology

4.1 Forecast Model: DA-BiGRU-CNN with Kink Subtraction

For each protocol $i \in \{\text{Aave}, \text{Compound}\}$ and decision time t, we observe the rolling window $x_i(t) = (r_i(\tau), u_i(\tau), \text{TVL}_i(\tau), \Delta\text{TVL}_i(\tau), \text{kink}_i)_{\tau=t-167}^{t}$ and predict $r_i(t+\Delta)$ at $\Delta = 12$ hours. The architecture decomposes additively:

- **Branch A — rate residual.** Target $\varepsilon_i(t) = r_i(t) - f_{\text{kink}_i}(u_i(t))$, where f_{kink_i} is the protocol-known piecewise-linear function; predicted by a 2-layer BiGRU with hidden size 64 over a 168-hour window of residuals plus the cross-protocol residual spread.
- **Branch B — utilization.** Target $\hat{u}_i(t+\Delta)$; predicted by a 2-layer BiGRU with hidden size 64 followed by a multi-scale 1-D CNN with kernels $\{3, 5, 7\}$, over a context window containing u, TVL, ΔTVL, gas, and time-of-day features per protocol.
- **Late fusion.** Reconstruct $\hat{r}_i(t+\Delta) = f_{\text{kink}_i}(\hat{u}_i(t+\Delta)) + \hat{\varepsilon}_i(t+\Delta)$, with a small MLP head correcting systematic bias in the additive reconstruction.

Figure 1 sketches the architecture. The kink function is a hard, exact, non-learned prior: the deterministic part of the rate is not estimated from data, freeing model capacity for the genuinely stochastic residual.

4.2 MCDM Scoring

We inherit the four-factor SAW aggregator of [19]. For protocol i at time t,

$$\mathrm{Score}_i(t) \;=\; w_1\, f_{\mathrm{APY}}(\hat{r}_i(t+\Delta)) + w_2\, f_{\mathrm{Risk}}(u_i(t)) + w_3\, f_{\mathrm{Cost}}(g(t)) + w_4\, f_{\mathrm{Stab}}(\Delta\mathrm{TVL}_i(t)), \quad (1)$$

with sub-factors

$$\begin{aligned} f_{\mathrm{APY}}(\hat{r}) &= \mathrm{clamp}(\hat{r}/\mathrm{APY}_{\max}, 0, 1), & f_{\mathrm{Risk}}(u) &= 1 - \mathrm{clamp}(u, 0, 1), \\ f_{\mathrm{Cost}}(g) &= 1 - \mathrm{clamp}(gG/g_{\max}, 0, 1), & f_{\mathrm{Stab}}(\Delta\mathrm{TVL}) &= 1 - \mathrm{clamp}(|\Delta\mathrm{TVL}|/0.30, 0, 1), \end{aligned}$$

default weights $(w_1, \ldots, w_4) = (0.40, 0.25, 0.20, 0.15)$, gas budget $G = 200{,}000$, $g_{\max} = 0.01$ ETH, and APY normalization $\mathrm{APY}_{\max} = 0.20$. The single critical departure from [19] is the substitution of the forecast $\hat{r}_i(t+\Delta)$ for the EMA-smoothed spot $S_i(t)$ in f_{APY}.

4.3 Decision Rule: Hysteresis, Cooldown, Gas Gate

The strategy executes the three-gate rebalance protocol in Listing 1: hysteresis $\theta = 0.05$ on the score delta, cooldown $\tau = 1$ h between actions, and a gas-cost gate that requires the expected uplift over the forecast horizon to exceed the realized gas spend.

```
def step(t):
    s_a = mcdm_score("AAVE", forecast_a, util_a, gas, dTVL_a)
    s_c = mcdm_score("COMPOUND", forecast_c, util_c, gas, dTVL_c)
    target, s_target = argmax((s_a, "AAVE"), (s_c, "COMPOUND"))
    if (s_target - s_current(t)) <= THETA: return []
    if (t - last_rebalance_time) < TAU_COOLDOWN: return []
    if expected_uplift * (horizon - t) * NOTIONAL
                                   < gas_cost(t): return []
    return [withdraw(current), supply(target)]
```

Listing 1: Predictive MCDM rebalance step (per hourly tick).

4.4 Loss Function: Composite Objective

Training minimizes the composite loss

$$\mathcal{L} = \alpha \cdot \mathrm{MSE}(r, \hat{r}) + \beta \cdot (1 - \mathrm{WPearson}(r, \hat{r})) + \gamma \cdot \mathrm{QuantileLoss}_{0.9}(r, \hat{r}), \quad (2)$$

with defaults $(\alpha, \beta, \gamma) = (0.4, 0.5, 0.1)$ tuned by validation grid search. The weighted-Pearson term carries over verbatim from [20], where it dominates MSE alone on allocation-relevant accuracy; the quantile term penalizes underprediction of large positive moves — exactly the moves the allocator most needs to anticipate.

5 Data

5.1 Sources

Aave V3 USDC reserve-rate history is pulled from the official Aave protocol-subgraphs deployment on TheGraph's decentralized network, subgraph ID `Cd2gEDVeqnjBn1hSeqFMitw8Q1-iiyV9FYUZkLNRcL87g` (entity `reserveParamsHistoryItems`, `liquidityRate` in annualized RAY scaling, 10^{27}). The Messari Compound V3 subgraph indexes per-collateral markets rather than the base supply market; USDC base-rate history is therefore reconstructed via batched `eth_call` on the Comet proxy `0xc3d688B66703497DAA19211EEdff47f25384cdc3` at historical block numbers, querying the `getUtilization()` and `getSupplyRate(uint256)` view functions. Gas prices come from Dune's `gas.gas_price` table at hourly median resolution; ETH/USD comes from

Table 1: Per-quarter cross-protocol spread structure on the full-coverage overlapping panel (n = 12,895 hours, 98.5% joint coverage). Spread is $\mathrm{APR}_{\mathrm{Aave}} - \mathrm{APR}_{\mathrm{Compound}}$ in percentage points; "Aave > C" is the share of hours on which Aave pays more. The 2025 Q3→Q4 jump from 39.9% to 56.3% (median spread crossing zero from −0.218 to +0.183 pp) is the empirical regime shift motivating Branch B of the forecaster: it is the first quarter in which Aave systematically pays more than Compound.

Quarter	n	spread median (pp)	spread std (pp)	Aave > C
2024 Q4	1455	−0.591	5.746	45.4%
2025 Q1	2160	−0.725	1.360	24.9%
2025 Q2	2184	−0.444	0.883	43.5%
2025 Q3	2208	−0.218	1.401	39.9%
2025 Q4	2208	+0.183	2.019	56.3%
2026 Q1	1960	−0.240	0.567	27.1%
2026 Q2	720	+0.063	3.605	61.3%

CoinGecko's free hourly close. Kink parameters per protocol per epoch are reconstructed from on-chain `ReservesUpdated` (Aave) and `Configurator` (Compound) events. The `fractal-defi` substrate of [12] is used as the loader and backtest harness.

5.2 Window and Resampling

Eighteen months: 1 November 2024 through 30 April 2026, a 13,104-hour grid. The window spans two macro micro-regimes (low-volatility H2 2024, normalizing 2025, and the present higher-volatility 2026) without overlapping any protocol re-deployment event. The Aave subgraph emits an event stream (median ~ 50 events per hour, $\sim 6.5 \times 10^5$ raw rows over the window) rather than uniform snapshots; the loader's `transform()` resamples the stream to a right-closed hourly grid via `df.resample("1H", label="right", closed="right").last()`, yielding **13,096 hourly Aave bars** (99.9% coverage). The Compound RPC-reconstruction yields **4,900 hourly bars** (37.4% raw coverage of the same window); the gap is filled incrementally as described in Section 5.4.

5.3 Empirical Spread Distribution

On the n = 12,895 overlapping hours where both protocols have observations (the 98.5% full-coverage joined panel; see Section 5.4), the cross-protocol supply-rate spread $s(t) = \mathrm{APR}_{\mathrm{Aave}}(t) - \mathrm{APR}_{\mathrm{Compound}}(t)$ has the following marginal statistics: median −0.232 pp, standard deviation 2.513 pp, range $[-26.79, +44.89]$ pp. The Aave-pays-more binary share is 40.7%; Compound pays more on the typical hour by 0.23 pp. The individual APR distributions are likewise broad: Aave USDC supply APR ranges $[1.54\%, 48.38\%]$ with median 3.82%; Compound USDC supply APR ranges $[2.85\%, 23.65\%]$ with median 4.46% (both quoted as annualized continuous-compound rates). An earlier sparse-sample analysis on the pre-top-up panel (n = 4,894, 37% Compound coverage) suggested a 2025 Q1→Q2 regime shift in the Aave-pays-more share; the 98.5% full-coverage panel reveals that that apparent jump was an artifact of cluster sampling around Nov–Dec 2024 and Jun–Aug 2025, and we report the corrected structure below.

Pooled marginals mask substantial structure. Table 1 partitions the overlapping panel by calendar quarter:

The direction-share jump from 39.9% in 2025 Q3 to 56.3% in 2025 Q4 — the first quarter where Aave systematically pays more than Compound, with median spread crossing zero from −0.218 pp to +0.183 pp — is a regime shift of the kind a dual-branch architecture with explicit cross-protocol residual input is designed to detect; the roughly ten-fold volatility variation from

5.75 pp (2024 Q4) to 0.57 pp (2026 Q1) also supplies natural anchors for the volatility-tertile ablation of Section 7.2. Crucially, the binary direction split is far from symmetric in either direction across the full panel (40.7% Aave-higher vs 59.3% Compound-higher), so a forecast-driven edge cannot come from a structural bias toward Aave — it must come from anticipating the crossovers, in keeping with the cointegration finding of [7] and against the equilibrium-impossibility framing of [8], which concerns the existence of a single fair rate, not short-horizon predictability of the realized spread.

5.4 Data Coverage and Limitations

Raw coverage is asymmetric at fetch time: Aave 99.9% (13,096/13,104 hourly bars), Compound 37.4% (4,900/13,104) on the initial Ankr-only fetch. The asymmetry is structural: the Messari Compound V3 subgraph indexes per-collateral markets and exposes `rates: None` on the base Comet market entity, so hourly snapshots must be reconstructed via `eth_call` on archive RPCs — and free-tier archive providers impose batch-size caps near 100 calls per request and per-day soft limits in the thousands. The fetcher accordingly tops up the panel in an idempotent append-mode across multiple free RPC providers (`publicnode` and Ankr) until coverage saturates; after top-up, Compound coverage rises to 98.5% on the joined panel (12,895 overlapping hours). For sequence-length-168 forecaster training, we restrict to dense subperiods in which both protocols have recent observations, and forward-fill missing Compound rows up to a 168-hour ceiling on the joined panel, marking longer gaps as a separate regime that is excluded from training and reported as edge cases.

5.5 Cleaning

Outliers (rates outside $[0, 50\%]$, single-step relative jumps $> 5\times$) are flagged as oracle glitches and interpolated. The sign-convention invariant borrowing_rate $\geq$ lending_rate is enforced as a test fixture (`tests/test_sign_convention.py`) at every timestamp on both protocols, not as a trust statement about library version. Z-score normalization uses training-split statistics only.

5.6 Splits

Strict block-number splits, to defend against subgraph-aggregation leakage:

- **Train** (10 months): 2024-11-01 – 2025-08-31, $\sim$6,700 hourly bars on both protocols.
- **Validation** (4 months): 2025-09-01 – 2025-12-31, $\sim$2,920 hourly bars on Aave but only $\sim$290 Compound-dense hours pre-top-up; forecaster validation noise on the composite loss is expected to reflect this.
- **Test** (4 months, held out): 2026-01-01 – 2026-04-30, $\sim$2,876 hourly bars on Aave, 50 Compound-real hours pre-top-up; the H1 evaluation on the full test window is Aave-driven in its initial form.

Residual Compound sparsity (the 1.5% of the joined panel still missing after top-up) means the H1 Sharpe-improvement claim on the full 18-month window is marginally degraded relative to a fully-dense panel; the evaluation protocol of Section 7 therefore reports both the full-window result and a restricted result on dense subperiods, and uses the 2025 Q4 regime-shift sub-window ($n = 2{,}208$ overlapping hours, 56.3% Aave-pays-more share) as an additional H1 anchor.

6 Calibration

6.1 Forecaster Hyperparameter Grid

A grid search is tracked in MLflow under selection criterion "validation weighted-Pearson plus directional accuracy $\geq 55\%$". The grid varies hidden dimension $\{32, 64, 96, 128\}$, BiGRU layers $\{1, 2\}$, CNN kernels $\{[3, 5, 7], [3, 5], [5]\}$, dropout $\{0.0, 0.1, 0.2\}$, learning rate $\{10^{-3}, 2\cdot 10^{-3}, 5\cdot 10^{-3}\}$, batch size $\{16, 32, 64\}$, sequence length $\{72, 168, 336\}$ hours, and a three-point grid over the loss weights (α, β, γ). Defaults are reported in Section 4.1 and Section 4.4; the full grid is logged in Section B.

6.2 Strategy Hyperparameter Grid

A second grid tunes hysteresis $\theta \in \{0.02, 0.05, 0.08, 0.10\}$, cooldown $\tau \in \{0.5, 1, 2, 6\}$ h, forecast horizon $\Delta \in \{6, 12, 24\}$ h, APY normalization $\text{APY}_{\text{max}} \in \{0.10, 0.15, 0.20, 0.25\}$, and MCDM weight vector under twenty Dirichlet samples plus the $(0.40, 0.25, 0.20, 0.15)$ baseline. Selection criterion: validation Sharpe under the turnover constraint turnover $< 2 \times$ best baseline turnover.

7 Backtesting Protocol

7.1 Execution Assumptions

Decision cadence is hourly; block time 12 s; supply and withdraw operations are protocol-level and incur no slippage; transaction cost is 200,000 gas $\times$ historical hourly median `gas_price_gwei` $\times 10^{-9} \times$ ETH/USD per rebalance. The withdraw-then-supply atomicity gap of approximately 12 s is modelled as zero yield on the migrating notional. Yield accrues continuously per hour at the active protocol's supply rate. The initial notional is \$1,000,000 USDC, large enough that gas costs are a meaningful but not dominant component of net return.

7.2 Regime Split

Post-hoc, we partition test hours into volatility tertiles (low/medium/high) of $\sigma(r)$ on a rolling 24-hour window, and report all headline metrics per tertile and pooled. Two event-flagged sub-windows are reported separately: in-window governance events changing kink parameters (breakpoint reports) and the out-of-window March 2023 USDC depeg, drawn from the [4] dataset and used as the OOD ablation (Table 3, row 14).

8 Results

This section is the placeholder skeleton for results that will arrive in Week 3 of the execution timeline (1–7 June 2026). The targets quoted are forward-looking under the pre-registered hypotheses of Section 4.

8.1 Headline: Predictive vs Reactive MCDM-EMA

The primary hypothesis is a Sharpe-ratio improvement of 0.2–0.5 of the predictive variant over the reactive MCDM-EMA baseline of [19], with statistical significance assessed via a 1000-replicate bootstrap of monthly Sharpe ratios. A single-panel equity-curve overlay (predictive vs reactive, \$1M notional, Jan–Apr 2026 test window) will form the title-page headline figure.

8.2 Per-Baseline Comparison

Table 2 lays out the five-strategy by six-metric grid. Entries marked TBD will be populated from the `backtest/run_baselines.py` and `run_main.py` artifacts.

Table 2: Per-baseline test-window comparison ($1M USDC, 4-month held-out test window 2026-01-01 – 2026-04-30). Numerical entries populated post-execution.

Strategy	Net APY	Sharpe	MaxDD	Calmar	Turnover	Gas ($)
A. Buy-and-hold Aave V3	TBD	TBD	TBD	TBD	TBD	TBD
B. APY-only greedy	TBD	TBD	TBD	TBD	TBD	TBD
C. Reactive MCDM-EMA	TBD	TBD	TBD	TBD	TBD	TBD
D. CIR-calibrated MCDM	TBD	TBD	TBD	TBD	TBD	TBD
E. Predictive MCDM (ours)	TBD	TBD	TBD	TBD	TBD	TBD

Table 3: Ablation matrix (15 rows). All metrics on test window. Numerical entries populated post-execution.

#	Ablation	Sharpe	DirAcc	WPearson	Turnover
1	No-forecast EMA baseline (=Baseline C)	TBD	TBD	TBD	TBD
2	Naive last-observation forecast	TBD	TBD	TBD	TBD
3	CIR-calibrated forecast (=Baseline D)	TBD	TBD	TBD	TBD
4	Markov-switching short rate (2-state)	TBD	TBD	TBD	TBD
5	CatBoost on hand-crafted features	TBD	TBD	TBD	TBD
6	Single-branch DA-GRU (no CNN, no decomp.)	TBD	TBD	TBD	TBD
7	Dual-branch *without* kink subtraction	TBD	TBD	TBD	TBD
8	Forecast with ranking loss only	TBD	TBD	TBD	TBD
9	MCDM equal weights vs tuned weights	TBD	TBD	TBD	TBD
10	Single-protocol benchmark (Aave/Compound only)	TBD	TBD	TBD	TBD
11	Greedy max-forecast vs TOPSIS-MCDM	TBD	TBD	TBD	TBD
12	Zero-gas vs realistic-gas	TBD	TBD	TBD	TBD
13	Sliding-window stability (14-d windows)	TBD	TBD	TBD	TBD
14	OOD test on USDC depeg week (Mar 2023)	TBD	TBD	TBD	TBD
15	Forecast horizon sweep (3/6/12/24 h)	TBD	TBD	TBD	TBD

8.3 Ablations

Table 3 lists the fifteen ablation rows. Rows 1–5 isolate the contribution of *any* forecast over EMA; rows 6–8 attribute remaining gains to architectural choices (single-branch vs dual-branch, kink subtraction, ranking loss); rows 9–11 examine the MCDM aggregator (equal weights, single-protocol fallback, greedy vs TOPSIS); rows 12–13 stress-test gas regime and sliding-window stability; rows 14–15 cover OOD-depeg and forecast horizon sweep.

8.4 Regime-Conditional Breakdown

The predictive-vs-reactive Sharpe delta and forecast direction accuracy are reported per volatility tertile. We pre-register the expectation that the forecast edge concentrates in medium and high tertiles, where cross-protocol dispersion is largest and the EMA baseline lags most.

8.5 Forecast Quality

Forecast-only diagnostics (R^2 on the rate level, direction accuracy on the binary "Aave > Compound at $t + 12\text{h}$" label, weighted-Pearson correlation, quantile-90 loss) are reported per protocol and pooled. The pre-registered targets are $R^2 \in [0.15, 0.40]$ on the rate level and directional accuracy $\in [55, 65]\%$.

9 Discussion

The real-data panel assembled prior to test-window exposure (Section 5) surfaces three empirical features that bear directly on the research questions of Section 4 and on the architectural choices defended in [20]. We discuss each in turn, then respond head-on to the Halpern–Pass–Saraf impossibility result.

9.1 Governance Dynamics as a Structured Noise Source

A snapshot of the Aave V3 USDC reserve at 2026-05-14 returns a post-kink slope $s_2 = 0.10$, sharply lower than the historical $s_2 \approx 0.60$ that prevailed for most of the training window. Cross-referencing on-chain `ReserveInterestRateStrategyChanged` events places the change in the Aave DAO's autumn 2025 risk-parameter cycle — temporally coincident with the 2025 Q3→Q4 cross-protocol regime shift documented in Section 9.2. The consequence for forecasting is striking: subtracting $f_{\text{kink}}(u; \theta_{\text{current}})$ from the realized supply rate on the full 13,096-bar panel using *current* parameters explains only 21.3% of supply-rate variance, with residual mean +186 bps and standard deviation 227 bps — far from the 60–80% that a naive application of the kink prior would suggest. Branch A of the proposed forecaster, whose target is precisely $\varepsilon_i(t) = r_i(t) - f_{\text{kink}}(u_i(t); \theta_{\text{current}})$, therefore absorbs a deterministic +186 bps offset that would otherwise be invisible to any architecture conditioning only on r and u. The governance shift, identified in Section 10.2 as a risk to the kink-subtraction design, turns into a signal the forecaster can be expected to internalize. A direct empirical prediction follows: ablation 7 (dual-branch *without* kink subtraction) should underperform the kinked variant most clearly on the pre-governance-change portion of the test window, and the gap should close on the post-change portion — a falsifiable architectural claim that the backtest will resolve. Time-varying kink parameters obtained by streaming the full `ReserveInterestRateStrategyChanged` event series into the loader are the natural follow-up extension.

9.2 Empirical Regime Structure Confirms Architectural Choice

Quarterly aggregation of the cross-protocol spread on the 12,895-hour full-coverage overlap reveals a regime shift between 2025 Q3 and 2025 Q4 that is inconsistent with stationary dynamics. The Aave-pays-more share moves from 39.9% in Q3 (still Compound-dominant; spread median −0.218 pp) to 56.3% in Q4 (Aave-dominant; spread median +0.183 pp), a sixteen percentage-point jump in conditional mean and the first sign flip of the median spread on the panel — the first quarter in which Aave systematically pays more than Compound. Per-quarter spread standard deviation across the panel ranges 0.57–5.75 pp; the Q3-to-Q4 shift in conditional mean is several multiples of the within-quarter scale. The panel additionally exhibits a strong volatility regime split: $\sigma = 5.75$ pp in 2024 Q4 (high-vol shock period) versus $\sigma = 0.57$ pp in 2026 Q1 (calm), an approximately tenfold variation that provides the natural anchor for the tertile-conditional reporting of Section 7.2. These findings constitute an empirical existence proof for the H2 architectural claim: regime structure that single-state models cannot capture is statistically present in the data, and a Markov-switching baseline along the lines of [9] (ablation 4) should show its largest gains over the EMA baseline precisely on the Q3–Q4 2025 transition window. The dual-branch DA-BiGRU-CNN architecture inherited from [20] is designed for exactly this kind of regime-dependent multivariate dynamics; the Q3-to-Q4 shift makes the architectural choice empirically motivated rather than merely transferred.

9.3 Response to Halpern–Pass–Saraf on Short-Horizon Predictability

[8] prove, by reduction of dynamic-repayment lending to barrier-option pricing, that no *fair* interest rate — in the sense of no-arbitrage under strategic borrowers who may delay repayment — exists for a realistic lending pool. They argue on the strength of this result that time-series

forecasting of pool rates is methodologically misplaced. The 12,895-hour full-coverage cross-protocol panel sharpens the response. The realized Aave-pays-more share is 40.7%, against 59.3% for Compound; the unconditional spread standard deviation is 2.51 pp, with extremes $[-26.79, +44.89]$ pp. A market in which cross-venue rate inequality reverses on roughly a 41/59 split, with material per-hour dispersion and a documented intra-panel regime shift (2025 Q3→Q4), is by construction *not* short-horizon efficient: persistent allocation edge is available to a forecaster that anticipates the crossovers ahead of the EMA-smoothed observer. The Halpern–Pass–Saraf theorem operates at the equilibrium-existence level and bounds the long-run fairness of any single rate; the predictive MCDM operates at the trade-timing level and exploits transient disequilibrium between two co-existing venues. The two claims are orthogonal: the impossibility of a fair equilibrium rate is fully compatible with, and in our panel empirically coexists with, an exploitable short-horizon structure. The 41/59 split with 2.51 pp standard deviation is, in our reading, a directly measured instance of the residual disequilibrium their result leaves untouched.

9.4 Limits of the LOB-to-DeFi Methodology Transfer

The transfer from [20] succeeds at the level of the dual-stream paradigm, the weighted-Pearson loss term, and the ranking-oriented evaluation. It is re-derived, not mechanically copied: LOB volume is unbounded and signed, while utilization is bounded and clamped; the kink prior has no LOB analogue and contributes the additional architectural lever that Section 9.1 shows to carry real explanatory mass. The principal residual risk is overfitting on a comparatively small (∼7k bar) training set; we mitigate via dropout, walk-forward cross-validation, and a deliberately compact (∼50k parameter) architecture. Under the null hypothesis H0 (Section 4), the dual-branch architecture neither helps nor hurts and the contribution reduces to the rigorous mainnet backtest of MCDM-EMA, the falsification of the LOB-to-DeFi transfer claim, and the open-source Extra+1/Extra+2 contributions of Section A — each of which stands on its own merits.

10 Conclusion and Future Work

10.1 Summary of Findings

We have specified, implemented, and pre-registered a forecast-driven MCDM allocator for USDC supply between Aave V3 and Compound V3, with a dual-branch DA-BiGRU-CNN forecaster exploiting the protocol-known kink function as a hard prior, on the `fractal-defi` substrate. The headline pre-registered targets are Sharpe improvement 0.2–0.5, $R^2 \in [0.15, 0.40]$, and directional accuracy $\in [55, 65]\%$ over a held-out 4-month test window. The null hypothesis is entertained explicitly and is structurally falsifiable by ablation 1.

10.2 Limitations

The 18-month window covers less than two full macro cycles; the two-protocol scope is deliberate but excludes Morpho, Spark, and cross-chain venues; the single underlying (USDC) is intentional but limits external validity. Gas-cost spikes can dominate forecast edge in adversarial regimes (ablation 12 quantifies the break-even gas). The forecast is conditional on neither protocol experiencing a solvency event during the test window — a historically valid but nontrivial assumption.

10.3 Future Work

Three extensions are immediate. First, Morpho and Spark integration would broaden the venue universe and test whether the predictive edge scales beyond the Aave–Compound pair. Second, a cross-chain extension (Aave on Arbitrum/Base) would expose the strategy to bridge friction and a richer execution-risk profile. Third, the dual-branch BiGRU–CNN architecture can be upgraded to attention-based backbones along the lines of [24] and [14]. A Panoptic-style options overlay on the supply position, hedging tail risk during the high-volatility tertile, is under active exploration.

A `fractal-defi` PR Artifacts

Extra+1: Compound V3 Lending Loader (with uniform utilization field). A new `CompoundV3LendingLoader` pulls hourly `marketHourlySnapshots` from the Messari subgraph, normalizes WAD scaling to the same per-second continuous-compound convention as the upstream Aave V3 loader, and emits a `LendingHistory` record. As a uniformity fix, both Aave and Compound `GlobalState` objects gain a `utilization` field computed from `totalLiquidity` and `totalCurrentVariableDebt` in the existing subgraph response. PR title: *"Add Compound V3 lending history loader (Messari subgraph) and uniform utilization field on lending `GlobalState`"*.

Extra+2: `BaseLendingAllocationStrategy`. A new abstract strategy template in `fractal/strategies/` with four overridable hooks (`compute_criteria_vector`, `aggregate_criteria`, `select_target`, `should_rebalance`). The proposed `PredictiveMCDMStrategy` is the first concrete subclass. PR title: *"Add `BaseLendingAllocationStrategy` for multi-lending-protocol allocators"*.

B Hyperparameter Grids

Full grids for both the forecaster (Section 6.1) and the strategy (Section 6.2) are logged in MLflow under `results/mlflow/`. The defaults reported in the body correspond to the configurations that maximize validation weighted-Pearson under the directional-accuracy floor and the turnover constraint, respectively.

C Reproducibility

Code is available at `https://github.com/SergeySolovyev/predictive-mcdm-defi`. Python dependencies are pinned in `requirements.txt` (`fractal-defi==1.3.2`, see ERRATA in `README.md`). Environment variables `THE_GRAPH_API_KEY`, `DUNE_API_KEY`, and `FRACTAL_DATA_PATH` configure the data fetchers. The full reproduction pipeline is the nine-step sequence in `README.md` §Quick start, ending in `latexmk -pdf main.tex`.

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